

Problem 8.13. Define $A_{\text{LBA}} = \{\langle M, w \rangle \mid M \text{ is an LBA that accepts input } w\}$. Show that A_{LBA} is PSPACE-complete.

Proof. To show that A_{LBA} is PSPACE-complete, we must show that it is in PSPACE and that all PSPACE problems are polynomial time reducible to it.

To show that A_{LBA} is PSPACE-complete, we can construct a PSPACE decider for A_{LBA} by using the algorithm given in Theorem 5.9¹. $\square$

¹An LBA operates in exactly n space. This solution applies to both DLBA and NLBA due to Savitch's Theorem.